

Experiment 2 – UNIX Commands & Shell Programming

```
(kali㉿kali)-[~]
└─$ date
Sat Oct 26 14:32:10 IST 2024

(kali㉿kali)-[~]
└─$ date "+%H-%M-%S"
14-32-15

(kali㉿kali)-[~]
└─$ cal 6 2026
      June 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30

(kali㉿kali)-[~]
└─$ echo Hello World
Hello World

(kali㉿kali)-[~]
└─$ echo "Line1 Line2 Line3"
Line1 Line2 Line3

(kali㉿kali)-[~]
└─$ who
kali      :0                2024-10-26 09:00 (:0)

(kali㉿kali)-[~]
└─$ who -H
NAME     LINE      TIME          COMMENT
kali     :0      2024-10-26 09:00 (:0)

(kali㉿kali)-[~]
└─$ who -b
          system boot  2024-10-26 09:00

(kali㉿kali)-[~]
└─$ who am i
kali     pts/0        2024-10-26 14:35 (:0)

(kali㉿kali)-[~]
└─$ tty
/dev/pts/0

(kali㉿kali)-[~]
└─$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
obase=2
10
1010
quit

(kali㉿kali)-[~]
└─$ man ls
LS(1)                                User Commands                                LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default).
    Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
    fied.
Manual page ls(1) line 1 (press h for help or q to quit)

(kali㉿kali)-[~]
└─$ ls
Desktop  Documents  Downloads  file1.txt  Music  Pictures  Public  Templates  Videos

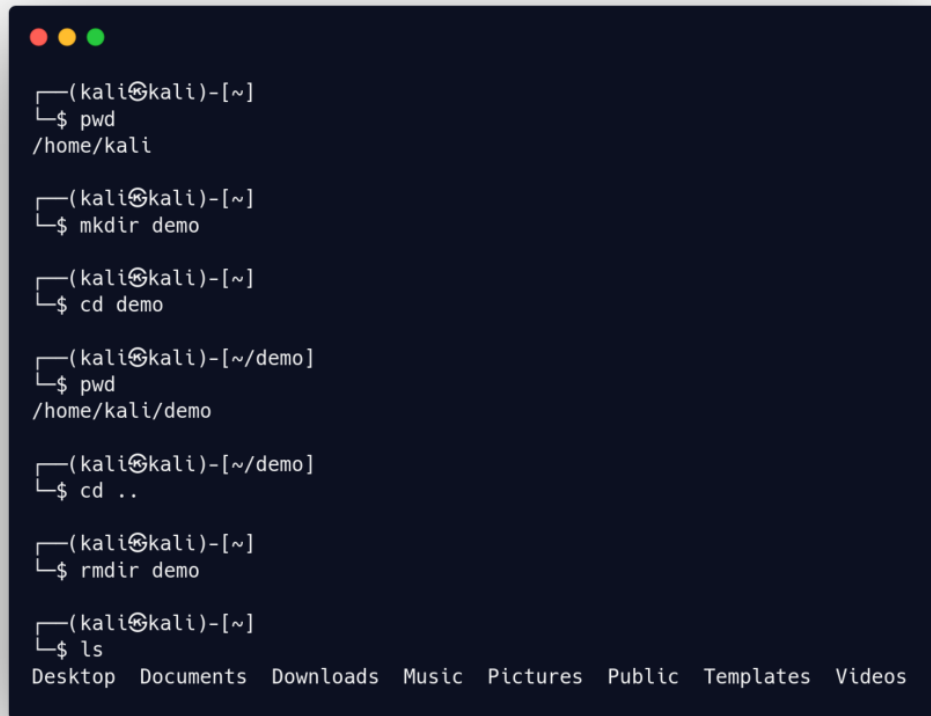
(kali㉿kali)-[~]
└─$ ls -a
.  .bashrc  Documents  .local  .profile
.. Desktop  Downloads  Music    .cache  Public

(kali㉿kali)-[~]
└─$ ls -l
total 4
-rw-r--r-- 1 kali kali   46 Oct 26 14:30 file1.txt
drwxr-xr-x 2 kali kali 4096 Oct 26 14:31 demo

(kali㉿kali)-[~]
└─$ ls -r
Videos  Templates  Public  Pictures  Music  Downloads  Documents  Desktop  file1.txt

(kali㉿kali)-[~]
└─$ ls -t
file1.txt  demo  Documents  Desktop  Downloads  Music  Pictures
```

2.Directory Related Commands

A terminal window with a dark blue background and white text. It shows a series of commands and their outputs in a Kali Linux environment. The prompt is (kali@kali)-[~]. The commands executed are: pwd (output: /home/kali), mkdir demo, cd demo, pwd (output: /home/kali/demo), cd .., rmdir demo, and ls (output: Desktop Documents Downloads Music Pictures Public Templates Videos).

```
(kali@kali)-[~]  
└─$ pwd  
/home/kali  
  
(kali@kali)-[~]  
└─$ mkdir demo  
  
(kali@kali)-[~]  
└─$ cd demo  
  
(kali@kali)-[~/demo]  
└─$ pwd  
/home/kali/demo  
  
(kali@kali)-[~/demo]  
└─$ cd ..  
  
(kali@kali)-[~]  
└─$ rmdir demo  
  
(kali@kali)-[~]  
└─$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

3.File related commands

```
(kali㉿kali)-[~]
└─$ cat > file1.txt
This is an OS Lab file.
It contains sample text.
^D

(kali㉿kali)-[~]
└─$ cat file1.txt
This is an OS Lab file.
It contains sample text.

(kali㉿kali)-[~]
└─$ cat file1.txt > file2.txt

(kali㉿kali)-[~]
└─$ cat file2.txt
This is an OS Lab file.
It contains sample text.

(kali㉿kali)-[~]
└─$ echo "Appending new line." >> file2.txt

(kali㉿kali)-[~]
└─$ cat file1.txt >> file2.txt

(kali㉿kali)-[~]
└─$ cat -n file1.txt
  1 This is an OS Lab file.
  2 It contains sample text.

(kali㉿kali)-[~]
└─$ cat > names.txt
Zara
Bob
Anna
^D

(kali㉿kali)-[~]
└─$ sort names.txt
Anna
Bob
Zara

(kali㉿kali)-[~]
└─$ cp file1.txt file3.txt

(kali㉿kali)-[~]
└─$ mv file3.txt renamed_file.txt

(kali㉿kali)-[~]
└─$ wc file1.txt
  2  9 46 file1.txt

(kali㉿kali)-[~]
└─$ wc -l file1.txt
 2 file1.txt

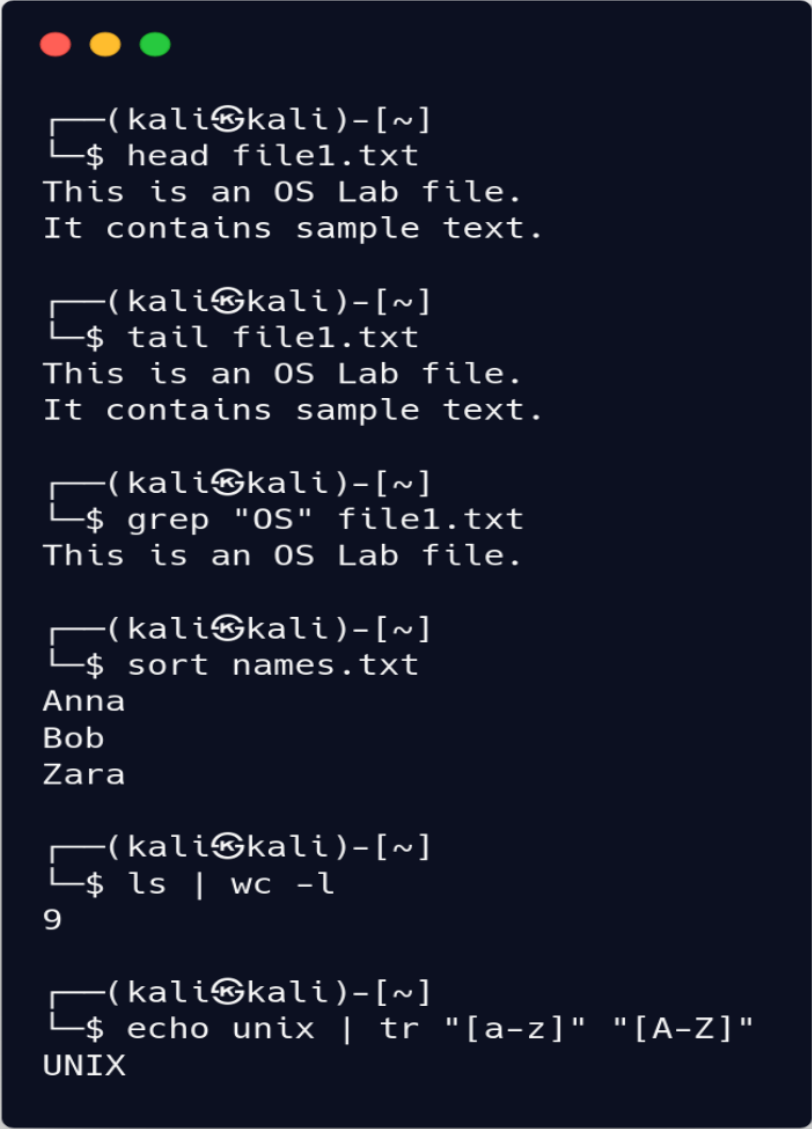
(kali㉿kali)-[~]
└─$ wc -w file1.txt
 9 file1.txt

(kali㉿kali)-[~]
└─$ wc -c file1.txt
46 file1.txt

(kali㉿kali)-[~]
└─$ rm renamed_file.txt

(kali㉿kali)-[~]
└─$ lp file1.txt
request id is HP-LaserJet-1 (1 file(s))
```

4.Filters and pipes



```
(kali㉿kali)-[~]
└─$ head file1.txt
This is an OS Lab file.
It contains sample text.

(kali㉿kali)-[~]
└─$ tail file1.txt
This is an OS Lab file.
It contains sample text.

(kali㉿kali)-[~]
└─$ grep "OS" file1.txt
This is an OS Lab file.

(kali㉿kali)-[~]
└─$ sort names.txt
Anna
Bob
Zara

(kali㉿kali)-[~]
└─$ ls | wc -l
9

(kali㉿kali)-[~]
└─$ echo unix | tr "[a-z]" "[A-Z]"
UNIX
```

5.Communication Through UNIX Commands

```
(kali㉿kali)-[~]
└─$ mesg y

(kali㉿kali)-[~]
└─$ write kali
Hey, this is a test message.
^D

(kali㉿kali)-[~]
└─$ wall "Server restart in 10 min"
Broadcast message from root@kali (pts/0) (Sat Oct 26 14:45:10 2024):

Server restart in 10 min

(kali㉿kali)-[~]
└─$ mail kali
Subject: Lab Manual
This is the OS lab record.
^D
EOT

(kali㉿kali)-[~]
└─$ mail
"/var/mail/kali": 1 message 1 new
>N 1 kali@kali Sat Oct 26 14:46 14/482 Lab Manual
? q
Held 1 message in /var/mail/kali. You have mail in /var/mail/kali
```

Shell programming outputs